

Vishnu Vardhan

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Profile:

Dynamic and results-driven Data Engineer with 5 years of experience specializing in building robust, scalable, and high-performance data pipelines. Proven expertise in designing and implementing solutions that optimize the flow of large-scale data across platforms. Adept in various programming languages, data processing, and storage technologies, including Python, SQL, Hadoop, and Spark. Skilled in collaborating with cross-functional teams to design, develop, and maintain end-to-end data solutions, providing actionable insights to drive business growth. Strong understanding of both data engineering principles and software engineering practices, delivering results with a focus on efficiency and scalability.

Technical Skills

- Big Data Technologies: Hadoop, Spark, Hive, Pig, Sqoop, Kafka
- Cloud Platforms: Azure, AWS, GCP
- Programming Languages: Java, Scala, Python
- Data Analytics: Data Warehousing, ETL, Data Mining
- Databases: SQL Server, Teradata, Redshift
- Data Visualization: Tableau, Power BI
- Agile Methodologies: Scrum, Kanban
- Operating Systems: Windows, Linux
- Data Security: Data Encryption, Access Control
- Machine Learning: Spark MLlib, scikit-learn
- Data Processing: MapReduce, Spark Streaming
- Data Storage: HDFS, Azure Blob Storage, AWS S3

Professional Experience:

Home Depot, Atlanta, GA

Jan 2024 – Present

Role: Data Engineer

Responsibilities:

- Architected and implemented scalable data pipelines using AWS Glue for efficient ETL (extract, transform, load) operations across large datasets.
- Designed data lakes on Amazon S3, utilizing AWS Lake Formation to govern and manage data security and access permissions.
- Leveraged AWS Redshift to design and optimize cloud-based data warehousing solutions, ensuring high performance and scalability.
- Built real-time data processing systems using AWS Kinesis to ingest and analyze streaming data for near-instantaneous insights.
- Automated ETL workflows with AWS Lambda, creating serverless solutions that improve scalability and reduce operational overhead.
- Used Amazon RDS for high availability and disaster recovery, optimizing database performance for transactional workloads.
- Integrated AWS S3 with Amazon Athena to query data directly from cloud storage, enabling low-cost and fast querying of large datasets.
- Utilized Amazon DynamoDB for building highly scalable, low-latency NoSQL databases for real-time applications.
- Built data integration workflows with AWS Step Functions, orchestrating AWS services such as Lambda, Glue, and S3 for end-to-end data processing.
- Experienced in designing and deployment of Hadoop cluster and different Big Data analytic tools including Pig, Hive, HBase, Oozie, Sqoop, Kafka, Spark with Cloudera distribution. Worked on Cloudera distribution and deployed on AWS EC2 Instances.
- Worked on integrating Apache Kafka with Spark Streaming process to consume data from external REST APIs and run custom functions.
- Involved in performance tuning of Spark jobs using Cache and using complete advantage of cluster environment.
- Developed Spark scripts by using Scala and python as per the requirement.

- Involved in running Hadoop streaming jobs to process terabytes of text data. Worked with different file formats such as Text, Sequence files, Avro, ORC and Parquet.
- Configured, supported, and maintained all network, firewall, storage, load balancers, operating systems, and software in AWS EC2.
- Managed and scaled data storage solutions using AWS S3 and Glacier, ensuring cost-efficient and secure storage for archival and backup.

Skills: Apache, Azure, Azure SQL Database, Blob, Cassandra, CI/CD, Data Factory, Docker, Elasticsearch, ETL, Factory, Git, HBase, HDFS, Java, Jenkins, JS, Kafka, Kubernetes, lake, MySQL, Oracle, PySpark, Python, Scala, Services, Snowflake, Spark, Spark Streaming, SQL, Sqoop, SSIS, Teradata.

M9 Consulting INC, Austin, TX

Jan 2023- Dec 2023

Role: Data Engineer

Responsibilities:

- Orchestrated data pipelines and transformations using AWS Data Pipeline, automating data movement between different AWS services and on-premises resources.
- Implemented security best practices with AWS IAM roles and policies to control access to AWS data resources, ensuring compliance and security.
- Deployed AWS EMR (Elastic MapReduce) clusters for distributed data processing, enabling big data analytics and Hadoop-based workloads.
- Optimized data processing workloads with AWS EC2 instances, ensuring the right instance types are used for cost and performance efficiency.
- Utilized AWS CloudWatch for monitoring and logging of data engineering workflows, ensuring visibility into pipeline health and performance.
- Designed and implemented AWS CloudFormation templates to automate the provisioning and deployment of AWS data resources.
- Utilized AWS Glue DataBrew for data wrangling and cleaning, automating data transformation processes and reducing manual effort.
- Built real-time event-driven architectures with AWS EventBridge, ensuring event-based triggers and workflows for data-driven applications.
- Integrated AWS SageMaker with data pipelines to deploy machine learning models and integrate predictive analytics into business processes.
- Managed and optimized AWS Redshift clusters using best practices, including workload management, query optimization, and data distribution strategies.

Skills: APIs, AWS, CI/CD, Docker, EMR, ETL, HBase, Java, Jenkins, Jira, Kafka, Kubernetes, lake, Lambda, Oracle, PySpark, Python, S3, Scala, Snowflake, Spark, SQL.

Zetaion Private Ltd, Bengaluru, India

April 2020 – July 2022

Role: AWS Data Engineer

Responsibilities:

- Streamlined data migration from on-premises databases to AWS Cloud using AWS Database Migration Service (DMS), ensuring minimal downtime.
- Developed and maintained serverless data processing pipelines using AWS Lambda and S3, optimizing cost and scalability.
- Implemented data quality and validation frameworks using AWS Glue and Lambda functions to ensure clean and reliable datasets.
- Leveraged Amazon QuickSight for business intelligence reporting and visualizations, integrating with AWS data services for real-time insights.
- Managed backup and recovery solutions using AWS Backup, ensuring data protection and business continuity for critical data assets.
- Built and optimized data lakes and data warehouses using AWS Redshift Spectrum to query data directly from Amazon S3.
- Configured and deployed data streaming solutions with AWS Kinesis Firehose to seamlessly load data into AWS analytics services for further processing.

- Automated the management of cloud resources and infrastructure using Terraform and Google Cloud Deployment Manager, ensuring repeatable, scalable, and compliant provisioning of resources.
- Integrated GCP's Identity and Access Management (IAM) services with data engineering workflows to enforce data security policies and provide fine-grained access controls across cloud resources.
- Integrated Google Cloud Data Fusion for seamless data integration and orchestration, enabling data pipelines to efficiently move and transform data across GCP services and external environments.
- Implemented robust disaster recovery strategies on GCP using Cloud Storage and backup solutions to ensure data durability and minimize downtime in mission-critical data engineering operations.
- Collaborated with cross-functional teams to define data requirements, ensuring the efficient design and deployment of data engineering solutions that aligned with business goals and technical constraints.
- Led the migration of on-premises data systems to GCP, utilizing Cloud Storage, BigQuery, and Data Transfer Services to ensure seamless transition and maintain business continuity.
- Conducted data quality assessments using tools like Data Quality Frameworks and Google Cloud's Data Catalog, ensuring that high-quality, trustworthy data was available for analytics and business decision-making.

Skills: AWS, CI/CD, Data Factory, Docker, EC2, ETL, Factory, Glue, Hive, Java, Jenkins, Kubernetes, Lambda, PL/SQL, Python, S3, Snowflake, Spark, Spark Streaming, SQL, Analytics, API, AZURE, Big Query, Bricks, Cluster, EMR, Lake, SDK, Services, Spark, SSIS, Tableau, VPC.